Friction: Test the Surfaces · Pupil Resources

Learning goal: I can compare friction on different run-out surfaces using repeated measurements and a controlled setup.

Arrival Questions (In your book)

1. What force opposes a block sliding along a floor?

2. If a surface has more friction, will the block roll a shorter or longer distance?

3. What unit do we use to measure stopping distance on a tabletop?

4. Can you ask an adult before starting a practical trial?

Starter Question

We change both the block and the run-out surface. Can stopping distance tell us which surface caused the difference?

A. No. Changing the block as well as the surface leaves more than one possible cause.

B. Yes. Any change always proves the surface was rougher.

C. Yes, if we only write down one number.

Method Strip

1. Set the low ramp and place Surface A flat at the bottom with the ruler zero at the ramp foot.

2. Hold the block at the release line and let go without pushing. STOP AND CHECK with your adult.

3. Measure the stopping distance to the front edge of the block in cm and record Trial 1 and Trial 2.

4. Replace Surface A with Surface B, keeping the ramp and block identical, and record two trials.

5. State whether you used practical or supplied data, and write which surface showed greater friction.

Exit Ticket

Which stopping distance shows more friction: 24 cm or 46 cm?

Name two things we kept the same to make the surface test fair.

Why should we keep a surprising result instead of rubbing it out?